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PROGRAM FalsePosition
  IMPLICIT NONE
  REAL :: a, b, p, fa, fb, fp, tol=1.0E-5
  INTEGER :: maxIter, iter

  ! Input
  WRITE(*,*) "Enter the initial guesses a and b:"
  READ(*,*) a, b

  WRITE(*,*) "Enter maximum iterations:"
  READ(*,*) maxIter

  ! Check initial values
  fa = f(a)
  fb = f(b)
  IF (fa * fb > 0.0) THEN
    WRITE(*,*) "Error: f(a) and f(b) must have opposite signs."
    STOP
  END IF

  ! Iterative False Position method
  iter = 0
  DO WHILE ( iter < maxIter)
    p = b - fb * (b - a) / (fb - fa)
    fp = f(p)

    ! Print table headings for the first iteration
    IF (iter == 0) THEN
      WRITE(*,*) "Iter      |a      |b      |p      |f(p)"
    END IF

    ! Print current iteration values
    WRITE(*, '(I5, 2F12.6, F12.6, F12.6)') iter, a, b, p, fp

    IF (ABS(p-b) < tol) EXIT ! Root found

    IF (fa * fp < 0.0) THEN
      b = p
      fb = fp
    ELSE
      a = p
      fa = fp
    END IF

    iter = iter + 1
  END DO

  ! Final result

  WRITE(*,*) "The root is:", p

  ! Function declaration
  CONTAINS
  Real FUNCTION f(x)
    REAL, INTENT(IN) :: x
    f = EXP(x) + 2**(-x)+2.0 * COS(x)- 6.0

  END FUNCTION

END PROGRAM FalsePosition

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